■ `Fit`

`Fit[`*data*`,` *funs*`,` *vars*`]` finds a least-squares fit to a list of data as a linear combination of the functions *funs* of variables *vars*.

The data can have the form `{{`x_1`,` y_1`,` …`,` f_1`},` `{`x_2`,` y_2`,` …`,` f_2`},` …`}`, where the number of coordinates *x*, *y*, . . . is equal to the number of variables in the list *vars*.

The data can also be of the form `{`f_1`,` f_2`,` …`}`, with a single coordinate assumed to take values `1`, `2`, `...`.

The argument *funs* can be any list of functions that depend only on the objects *vars*.

`Fit[{`f_1`,` f_2`,` …`},` `{1, x, x^2},` `x]` gives a quadratic fit to a sequence of values f_i. The result is of the form a_0 `+` a_1 `x` `+` a_2 `x^2`, where the a_i are real numbers. The successive values of `x` needed to obtain the f_i are assumed to be `1`, `2`, ... ■ `Fit[{{`x_1`,` f_1`},` `{`x_2`,` f_2`},` …`},` `{1, x, x^2},` `x]` does a quadratic fit, assuming a sequence of `x` values x_i. ■ `Fit[{{`x_1`,` y_1`,` f_1`},` …`},` `{1, x, y},` `{x, y}]` finds a fit of the form a_0 `+` a_1 `x` `+` a_2 `y`. ■ `Fit[`*data*`,` `{{`fn_1`,` fn_2`,` …`},` `{`fd_1`,` …`}},` *vars*`]` fits the data to rational function of the form `(`an_1 fn_1 `+` an_2 fn_2 `+` …`)` `/` `(`ad_1 fd_1 `+` …`)`. ■ `Fit` always finds the linear combination of the functions in the list *forms* that minimizes the sum of the squares of deviations from the values f_i. ■ See page 461. ■ See also: `Solve`, `SingularValues`, `FindMinimum`.

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